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## EDUCATION

- 2000 – 2004: **Ph.D. in Physics**, University of La Plata (Argentina). Thesis: *Propagation of High Energy Galactic Cosmic Rays*, passed with the maximum score (10).
- 1993 – 1999: *Licenciado* in Physics (approx. equivalent to B.Sc. + M.Sc.), University of La Plata (Argentina). Cumulative GPA 9.57 (out of 10 max.).

## PROFESSIONAL SKILLS & EXPERTISE

- Highly productive and creative with **>110** peer-reviewed research articles (see full list below).
- Very strong problem solving skills, versatility, and ability to learn quickly.
- Excellent team working ability and strong communication and presentation skills developed in different research groups in the USA, Europe, and South America.
- 2 yrs Technical Services Engineer with leading EHR vendor Epic Systems Corp.
- Sun Certified Programmer for Java 6 (2008); long programming experience with Fortran77, R, Perl, Python; familiar with Unix, Mac OS X, Windows, and many numerical applications.
- Extensive experience in computational and theoretical modeling: Monte Carlo method, partial differential equations, complex network theory, data mining, SVD/PCA/tSNE, regularized generalized linear modeling, random forests, SVM, AI.

## RESEARCH INTERESTS

- Computational biology and bioinformatics applied to systems biology, single-cell biology, next-generation sequencing, high-multiplex proteomics, and new biotechnologies.
- Computational/mathematical modeling, AI, machine learning, biomedical database integration, data mining, complex network science, statistical analysis, data science.

## CAREER HISTORY

- Since 2014 (ongoing): **Staff Scientist** at the National Institutes of Health with appointments at the Trans-NIH Center for Human Immunology (2014-2018), CCR/NCI (2018-2021) and NIA (2021 to date).
- 2011- 2014: Research Associate at University of Maryland (College Park & Baltimore, MD).
- 2009 – 2011: EHR Technical Services Engineer at Epic Systems Corp (Verona, WI).
- 2006 – 2008: Postdoctoral Research Associate at the Center for Complex Network Research (Barabási Lab) at Notre Dame (South Bend, IN) and Northeastern University (Boston, MA).
- 2006: Visiting Scientist at University of New Mexico (Albuquerque, NM).
- 2004 - 2005: Postdoctoral Fellow at the Abdus Salam ICTP (Trieste, Italy).
- 2004: Guest Scientist at the Theory Division of Fermilab (Batavia, IL).
- 2000 – 2003: Ph.D. Fellow of CONICET (Argentina).
- 1996 – 2003: Teaching Assistant at the Physics Dept, University of La Plata (Argentina).

## PUBLICATIONS IN PEER-REVIEWED JOURNALS (\*equal contributions)

- 117) Valli G, Sarto F, Negro F, Monti E, Sirago G, Paganini M, Zampieri S, Franchi MV, Casolo A, Candia J, Ferrucci L, Narici MV, De Vito G, *Changes in motor unit conduction velocity after unilateral lower limb suspension and active recovery correlate with muscle ion channel gene expression*, bioRxiv 2025.02.26.640329 (under review).
- 116) Candia J, Fantoni G, Delgado-Peraza F, Shehadeh N, Tanaka T, Moaddel R, Walker KA, Ferrucci L, *SomaModules: a pathway enrichment approach tailored to SomaScan data*. Journal of Proteome Research, doi:10.1021/acs.jproteome.4c01114 (2025).
- 115) Premathilake HU, Mazucanti CH, Yao Q, O'Connell JF, Vegesna N, Tsitsipatis D, Weller C, Lam K-W G, Candia J, Fan J, De S, Sen P, Egan JM, Doyle ME, *Pig Taste Cell-Derived Organoids Synthesize Insulin* (to appear in Endocrinology).
- 114) Fabbri E, Candia J, Tanaka T, Moore AZ, Muratori P, Brugiavini A, Calderon Larrañaga A, Vetrano DL, Fratiglioni L, Crimmins E, Faul J, Langa KM, Weir D, Ferrucci L, *Dynamics of Multimorbidity, Health Expectancy and Survival in Middle Aged and Older Individuals* (to appear in Journal of Gerontology: Medical Sciences).
- 113) Ferrucci L, Herman A, Candia J, Wilson III D, Donega S, Picard M, *Inflammation and Stress Responses in Cardiovascular Disease* (to appear in Nature Reviews Cardiology).
- 112) Sathyan S, Liu F, Tanaka T, Ferrucci L, Ayers E, Windham G, Weiss E, Candia J, Coresh J, Barzilai N, Milman S, Walker KA, Verghese J, *A frailty-based plasma proteomic signature capturing overall health and well-being in older adults* (to appear in Aging Cell).
- 111) Chia R, Moaddel R, Kwan JY, et al, *Discovery of a plasma proteomics-based candidate biomarker panel predictive of amyotrophic lateral sclerosis* (to appear in Nature Medicine).
- 110) Herzog C, Poganik JR, Barzilai N, Basisty N, Beerman I, Belsky DW, de Cabo R, Candia J, Faghri F, Horvath S, Maier AB, Perez V, Sen P, Moqri M, Gladyshev VN, Ferrucci L, *Biomarkers of Aging-NIA Joint Symposium 2024: New Insights into Aging Biomarkers*. Aging Cell 24:e70124 (2025).
- 109) Moaddel R, Sanehira C, Keyes G, Cui C-Y, Ahmadkhania R, Candia J, Price NL, Eckroth S, Middleton B, Khadeer M, Mazucanti CH, McDevitt RA, Gorospe M, de Cabo R, Egan JM, Ramsden CE, Ferrucci L, *Senolytic effects of a modified Gingerenone A*. npj Aging 11:45 (2025).
- 108) Duggan MR, Yang S, Gomez GT, Cui Y, Capuano AW, Chen J, Yang Z, Wen J, Erus G, Drouin SM, Zweibaum D, Tian Q, Candia J, Bilgel M, Lewis A, Moghekar A, Ashton NJ, Kac PR, Karikari TK, Blennow K, Zetterberg H, Maher BS, Spira AP, Dumitrescu L, Hohman TJ, Gottesman RF, Davatzikos C, Bennett DA, Coresh J, Ferrucci L, Resnick SM, Yolken R, Walker KA, *Proteomic signatures of corona and herpes viral antibodies identify IGDCC4 as a mediator of neurodegeneration*. Science Advances 11:eadt7176 (2025).
- 107) Ma L, Hung MH, Mehrabadi FR, Wang L, Li Q, Forgues M, Wang KC, Budhu A, Candia J, Chaisaingmongkol J, Rabibhadana S, Pupacdi B, Ruchirawat M, Wang XW, *Beneficial infections of the enterovirus genus in patients with liver cancer*, Gut, doi:10.1136/gutjnl-2024-334681 (2025).
- 106) Tanaka T, Rosano C, Huang X, Tian Q, Landman BA, Moore AZ, Miljkovic I, Perry A, Khan S, Kalhan R, Carr JJ, Terry JG, Yaffe K, Walker K, Candia J, Ferrucci L, *Plasma proteomic analysis of intermuscular fat links muscle integrity with processing speed in older adults*, Alzheimer's & Dementia 21:e70261 (2025).
- 105) Candia J, Fantoni G, Moaddel R, Delgado-Peraza F, Shehadeh N, Tanaka T, Ferrucci L, *Effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using the SomaScan and Olink assays*. Journal of Proteome Research 24:2517 (2025).
- 104) Olinger B, Banarjee R, Dey A, Tsitsipatis D, Tanaka T, Ram A, Nyunt T, Daya G, Peng Z, Cui L, Candia J, Simonsick EM, Gorospe M, Walker KA, Ferrucci L, Basisty N, *The secretome of senescent monocytes predicts age-related clinical outcomes in humans*. Nature Aging, doi:10.1038/s43587-025-00877-3.

- 103)** Moaddel R\*, Candia J\*, Ubaida-Mohien C\*, Tanaka T, Moore AZ, Zhu M, Fantoni G, Church S, D'Agostino J, Fan J, Shehadeh N, De S, Lehrmann E, Kaileh M, Simonsick E, Sen R, Egan JM, Ferrucci L, *Healthy Aging Metabolomic and Proteomic Signatures Across Multiple Physiological Compartments*. *Aging Cell* 24:e70014 (2025).
- 102)** Ubaida-Mohien C, Moaddel R, Spendiff S, MacMillan NJ, Filion ME, Morais JA, Candia J, Fitzgerald LF, Taivassalo T, Coen PM, Ferrucci L, Hepple RT, *Serum Proteomic and Metabolomic Signatures of High versus Low Physical Function in Octogenarians*. *Aging Cell* 24:e70002 (2025).
- 101)** Franchi MV, Candia J, Sarto F, Sirago G, Valli G, Paganini M, Hartnell L, Giacomello E, Toniolo L, Monti E, Nogara L, Moro T, Paoli A, Murgia M, Brocca L, Pellegrino MA, Grassi B, Bottinelli R, De Vito G, Ferrucci L, Narici MV, *Previous short-term disuse dictates muscle gene expression and physiological adaptations to subsequent resistance exercise*. DOI:10.1113/JP287003. *Journal of Physiology* (2025).
- 100)** Candia J, Fantoni G, Delgado-Peraza F, Shehadeh N, Tanaka T, Moaddel R, Walker KA, Ferrucci L, *Variability of 7K and 11K SomaScan plasma proteomics assays*. *Journal of Proteome Research* 23:5531 (2024).
- 99)** Martinez-Romero J, Fernandez ME, Bernier M, Price NL, Mueller W, Candia J, Camandola S, Meirelles O, Hu Y-H, Li Z, Asefa N, Deighan A, Vieira Ligo Teixeira C, Palliyaguru DL, Serrano C, Escobar-Velasquez N, Dickinson S, Shiroma EJ, Ferrucci L, Churchill GA, Allison DB, Launer LJ, de Cabo R, SLAM Investigators, *A hematology-based clock derived from the Study of Longitudinal Aging in Mice to estimate biological age*. *Nature Aging* 4:1882 (2024).
- 98)** Apps R\*, Biancotto A\*, Candia J\*, Kotliarov K\*, Perl S, Cheung F, Farmer R, Mulè MP, Rachmaninoff N, Chen J, Martins AJ, Shi R, Zhou H, Bansal N, Schum P, Olnes MJ, Milanez-Almeida P, Han KL, Sellers B, Cortese M, Hagan T, Roupheal N, Pulendran B, King L, Manischewitz J, Khurana S, Golding H, van der Most RG, Dickler HB, Germain RN, Schwartzberg PL, Tsang JS, *Acute and persistent responses after H5N1 vaccination in humans*. *Cell Reports* 43:114706 (2024).
- 97)** Duggan MR, Peng Z, Sipilä PN, Lindbohm JV, Chen J, Lu Y, Davatzikos C, Erus G, Hohman TJ, Andrews SJ, Candia J, Tanaka T, Joynes CM, Alvarado CX, Nalls MA, Cordon J, Daya GN, An Y, Lewis A, Moghekar A, Palta P, Coresh J, Ferrucci L, Kivimäki M, Walker KA, *Proteomics identifies potential immunological drivers of post-infection brain atrophy and cognitive decline*. *Nature Aging* 4:1263 (2024).
- 96)** Duggan MR, Gomez GT, Joynes CM, Bilgel M, Chen J, Fattorelli N, Hohman TJ, Mancuso R, Cordon J, Castellano T, Koran MEI, Candia J, Lewis A, Moghekar A, Ashton NJ, Kac PR, Karikari TK, Blennow K, Zetterberg H, Martinez-Muriana A, De Strooper B, Thambisetty M, Ferrucci L, Gottesman RF, Coresh J, Resnick SM, Walker KA, *Proteome-wide analysis identifies plasma immune regulators of amyloid-beta progression*. *Brain, Behavior, and Immunity* 120:604 (2024).
- 95)** Candia J, Ferrucci L, *Assessment of Gene Set Enrichment Analysis using curated RNA-seq-based benchmarks*. *PLoS ONE* 19:e0302696 (2024).
- 94)** Kacar Z, Slud E, Levy D, Candia J, Budhu A, Forgues M, Wu X, Raziuddin A, Tran B, Shetty J, Pomyen Y, Chaisaingmongkol J, Rabibhadana S, Pupacdi B, Bhudhisawasdi V, Lertprasertsuke N, Auewarakul C, Sangrajrang S, Mahidol C, Ruchirawat M, Wang XW. *Characterization of tumor evolution by functional clonality and phylogenetics in hepatocellular carcinoma*. *Communications Biology* 7:383 (2024).
- 93)** Roberts JA, Basu-Roy S, Shin J, Varma VR, Williamson A, Blackshear C, Griswold ME, Candia J, Elango P, Karikinneth AC, Tanaka T, Ferrucci L, Thambisetty M, *Serum proteomic signatures of common health outcomes among older adults*. *Gerontology* 70:269 (2024).
- 92)** Moaddel R, Ubaida-Mohien C, Tanaka T, Tian Q, Candia J, Moore AZ, Lovett J, Fantoni G, Shehadeh N, Turek L, Collingham V, Kaileh M, Chia CW, Sen R, Egan JM, Ferrucci L, *Cross-sectional analysis of healthy individuals across decades: aging signatures across multiple physiological compartments*. *Aging Cell* 23:e13902 (2024).
- 91)** Dark H, Paterson C, Daya GN, Peng Z, Duggan MR, Bilgel M, An Y, Moghekar A, Davatzikos C, Resnick SM, Loupy K, Simpson M, Candia J, Mosley T, Coresh J, Palta P, Ferrucci L, Shapiro A, Williams

SA, Walker KA, *Proteomic indicators of health predict Alzheimer's disease biomarker levels and dementia risk*. Annals of Neurology 95:260 (2024).

**90)** Panigrahi G, Candia J, Dorsey TH, Tang W, Ohara Y, Byun JS, Minas TZ, Zhang A, Ajao A, Cellini A, Yfantis HG, Flis AL, Mann D, Ioffe O, Wang XW, Liu H, Loffredo CA, Napoles AM, Ambs S, *Diabetes-associated breast cancer is molecularly distinct and shows a DNA damage repair deficiency*. JCI Insight 8:e170105 (2023).

**89)** Huth T, Dreher EC, Lemke S, Fritzsche S, Sugiyanto RN, Castven D, Ibberson D, Sticht C, Eiteneuer E, Jauch A, Pusch S, Albrecht T, Goeppert B, Candia J, Wang XW, Ji J, Marquardt JU, Nahnsen S, Schirmacher P, Roessler S, *Chromosome 8p-engineering reveals increased metastatic potential targetable by patient-specific synthetic lethality in liver cancer*. Science Advances 9:eadh1442 (2023).

**88)** Tin A, Fohner AE, Yang Q, Brody JA, Davies G, Yao J, Liu D, Caro I, Lindbohm JV, Duggan MR, Meirelles O, Harris SE, Gudmundsdottir V, Taylor AM, Henry A, Beiser AS, Shojaie A, Coors A, Fitzpatrick AL, Langenberg C, Satizabal CL, Sitlani CM, Wheeler E, Tucker-Drob EM, Bressler J, Coresh J, Bis JC, Candia J, Jennings LL, Pietzner M, Lathrop M, Lopez OL, Redmond P, Gerszten RE, Rich SS, Heckbert SR, Austin TR, Hughes TM, Tanaka T, Emilsson V, Vasan RS, Guo X, Zhu Y, Tzourio C, Rotter JI, Walker KA, Ferrucci L, Kivimäki M, Breteler MMB, Cox SR, Debette S, Mosley TH, Gudnason VG, Launer LJ, Psaty BM, Seshadri S, Fornage M, *Identification of circulating proteins associated with general cognitive function among middle-aged and older adults*. Communications Biology 6:1117 (2023).

**87)** Cordon J, Duggan MR, Gomez GT, Pucha A, Peng Z, Dark HE, Davatzikos C, Erus G, Lewis A, Moghekar A, Candia J, Ferrucci L, Kapogiannis D, Walker KA, *Identification of clinically relevant brain endothelial cell biomarkers in plasma*. Stroke 54:2853 (2023).

**86)** Roy R, Kuo P-L, Candia J, Sarantopoulou D, Ubaida-Mohien C, Hernandez D, Kaileh M, Arepalli S, Singh A, Bektas A, Kim J, Moore AZ, Tanaka T, McKelvey J, Zukley L, Nguyen C, Wallace T, Dunn C, Wood W, Piao Y, Coletta C, De S, Sen JM, Weng N, Sen R, Ferrucci L, *Epigenetic signature of human immune aging in the GESTALT study*. eLife 12:e86136 (2023).

**85)** Das JK, Banskota N, Candia J, Griswold ME, Orenduff M, de Cabo R, Corcoran DL, Das SK, De S, Huffman KM, Kraus VB, Kraus WE, Martin CK, Racette SB, Redman LM, Schilling B, Belsky DW, Ferrucci L, *Calorie restriction modulates the transcription of genes related to stress response and longevity in human muscle: The CALERIE study*. Aging Cell 22:e13963 (2023).

**84)** Minas TZ, Lord BD, Zhang AL, Candia J, Dorsey TH, Baker FS, Tang W, Bailey-Whyte M, Smith CJ, Obadi OM, Ajao A, Jordan SV, Tettey Y, Biritwum RB, Adjei AA, Mensah JE, Hoover RN, Hsing AW, Liu J, Loffredo CA, Yates C, Cook MB, Ambs S, *Circulating trans fatty acids are associated with prostate cancer in Ghanaian and American men*. Nature Communications 14:4322 (2023).

**83)** Roberts J, Varma VR, Candia J, Tanaka T, Ferrucci L, Bennett DA, Thambisetty M, *Unbiased proteomics and multivariable regularized regression techniques identify SMOC1, NOG, APCS, and NTN1 in an Alzheimer's disease brain proteomic signature*. npj Aging 9:18 (2023).

**82)** Osawa Y, Candia J, Abe Y, Tajima T, Oguma Y, Arai Y, *Plasma amino acid signature for sarcopenic phenotypes in community-dwelling octogenarians: Results from the Kawasaki Aging Wellbeing Project*. Experimental Gerontology 178:112230 (2023).

**81)** Ferrucci L, Candia J, Ubaida-Mohien C, Lyashkov A, Banskota N, Leeuwenburgh C, Wohlgemuth S, Guralnik JM, Kaileh M, Zhang D, Sufit R, De S, Gorospe M, Munk R, Peterson CA, McDermott MM, *Transcriptomic and proteomic of gastrocnemius muscle in peripheral artery disease*. Circulation Research 132:1428 (2023) [Editorial: Jain I, Oropeza BP, Huang NF, *Multomics Analyses of Peripheral Artery Disease Muscle Biopsies*. Circulation Research 132:1444 (2023).]

**80)** Sirago G, Candia J, Franchi MV, Sarto F, Monti E, Toniolo L, Reggiani C, Giacomello E, Zampieri S, Hartnell LM, De Vito G, Sandri M, Ferrucci L, Narici, MV, *Upregulation of sarcolemmal hemichannels and inflammatory transcripts with neuromuscular junction instability during lower limb unloading in humans*. Biology 12:431 (2023).

- 79) Tanaka T, Talegawkar SA, Jin Y, [Candia J](#), Fantoni G, Bandinelli S, Ferrucci L, *Proteomic mediators of overall cardiovascular health on all-cause mortality*. *Nutrients* 15:781 (2023).
- 78) Cheung F, Apps R, Dropulic D, Kotliarov Y, Chen J, Jordan T, Langweiler M, [Candia J](#), Biancotto A, Han KL, Rachmaninoff N, Pietz H, Wang K, Tsang JS, Jeffrey I. Cohen, *Sex and prior exposure jointly shape innate immune responses to a live herpesvirus vaccine*. *eLife* 12:e80652 (2023).
- 77) [Candia J](#), Daya GN, Tanaka T, Ferrucci L, Walker KA, *Assessment of Variability in the Plasma 7k SomaScan Proteomics Assay*. *Scientific Reports* 12:17147 (2022).
- 76) Sarto F, Stashuk DW, Franchi MV, Monti E, Zampieri S, Valli G, Sirago G, [Candia J](#), Hartnell LM, Paganini M, McPhee JS, De Vito G, Ferrucci L, Reggiani C, Narici MV, *Effects of short-term unloading and active recovery on human motor unit properties, neuromuscular junction transmission and transcriptomic profile*. *Journal of Physiology* 600:4731 (2022).
- 75) Wang L\*, [Candia J](#)\*, Ma L\*, Zhao Y\*, Imberti L, Sottini A, Quiros-Roldan E, Dobbs K, Burbelo PD, Cohen JI, Delmonte OM, Forgues M, Liu H, Matthews HF, Shaw E, Stack MA, Weber S, Zhang Y, Lisco A, Sereti I, Su HC, Notarangelo LD, Wang XW, *Serological responses to human virome define clinical outcomes of Italian patients infected with SARS-CoV-2*. *Int. J. Biol. Sci.* 18:5591 (2022).
- 74) Tanaka T, Talegawkar SA, Jin Y, [Candia J](#), Tian Q, Moaddel R, Simonsick EM, Ferrucci L, *Metabolomic profile of different dietary patterns and their association with frailty index in community-dwelling older men and women*. *Nutrients* 14:2237 (2022).
- 73) Pat N, Wang Y, Bartonicek A, [Candia J](#), Stringaris A, *Explainable Machine Learning Approach to Predict and Explain the Relationship between Task-based fMRI and Individual Differences in Cognition*. *Cerebral Cortex* 1-22 (2022).
- 72) Osawa Y, Tanaka T, Semba RD, Fantoni G, Moaddel R, [Candia J](#), Simonsick EM, Bandinelli S, Ferrucci L. *Plasma growth and differentiation factor 15 predict longitudinal changes in bone parameters in women, but not in men*. *J Gerontol A Biol Sci Med Sci* XX,1-8 (2022).
- 71) Minas TZ\*, [Candia J](#)\*, Dorsey TH, Baker F, Tang W, Kiely M, Smith CJ, Zhang AL, Jordan SV, Obadi OM, Ajao A, Tettey Y, Biritwum RB, Adjei AA, Mensah JE, Hoover RN, Jenkins FJ, Kittles R, Hsing AW, Wang XW, Loffredo CA, Yates C, Cook MB, Ambis S. *Serum proteomics links suppression of tumor immunity to ancestry and lethal prostate cancer*. *Nature Communications* 13:1759 (2022).
- 70) Osawa Y, Tanaka T, Semba RD, Fantoni G, Moaddel R, [Candia J](#), Simonsick EM, Bandinelli S, Ferrucci L. *Proteins in the pathway from high red blood cell width distribution to all-cause mortality*. *EBioMedicine* 76:103816 (2022).
- 69) Yamaguchi Y, Zampino M, Tanaka T, Bandinelli S, Moaddel R, Fantoni G, [Candia J](#), Ferrucci L, Semba RD. *The plasma proteome fingerprint associated with circulating carotenoids and retinol in older adults*. *Journal of Nutrition* 152:40 (2022).
- 68) Roberts JA, Varma VR, An Y, Varma S, [Candia J](#), Fantoni G, Tiwari V, Anerillas C, Williamson A, Saito A, Loeffler T, Schilcher I, Moaddel R, Khadeer M, Lovett J, Tanaka T, Pletnikova O, Troncoso JC, Bennett DA, Albert MS, Yu K, Niu M, Haroutunian V, Zhang B, Peng J, Croteau DL, Resnick SM, Gorospe M, Bohr VA, Ferrucci L, Thambisetty M. *A Brain Proteomic Signature of Incipient Alzheimer's Disease in Young APOE ε4 Carriers Identifies Novel Drug Targets*. *Science Advances* 7:eabi8178 (2021).
- 67) Ma L, Wang L, Khatib SA, Chang C-W, Heinrich S, Dominguez DA, Forgues M, [Candia J](#), Hernandez MO, Kelly M, Zhao Y, Tran B, Hernandez JM, Davis JL, Kleiner DE, Wood BJ, Greten TF, Wang XW. *Single-cell atlas of tumor cell evolution in response to therapy in hepatocellular carcinoma and intrahepatic cholangiocarcinoma*. *Journal of Hepatology* 75:1397 (2021).
- 66) Landino K, Tanaka T, Fantoni G, [Candia J](#), Bandinelli S, Ferrucci L. *Characterization of the plasma proteomic profile of frailty phenotype*. *GeroScience* 43:1029 (2021).
- 65) Han K, Singh K, Rodman MJ, Hassanzadeh S, Baumer Y, Huffstutler RD, Chen J, [Candia J](#), Cheung F, Stagliano KER, Pirooznia M, Powell-Wiley TM, Sack MN. *Identification and Validation of Nutrient State-Dependent Serum Protein Mediators of Human CD4+ T Cell Responsiveness*. *Nutrients* 13:1492 (2021).

- 64) Matsuda K, Migueles SA, Huang J, Bolkhovitinov L, Stuccio S, Griesman T, Pullano AA, Kang BH, Ishida E, Zimmerman M, Kashyap N, Martins KM, Stadlbauer D, Pederson J, Patamawenu A, Wright N, Shofner T, Evans S, Liang CJ, Candia J, Biancotto A, Fantoni G, Poole A, Smith J, Alexander J, Gurwith M, Krammer F, Connors M. *A Replication Competent Adenovirus-Vectored Influenza Vaccine Induces Durable Systemic and Mucosal Immunity*. *Journal of Clinical Investigation* 131:e140794 (2021).
- 63) Zampino M, Tanaka T, Ubaida-Mohien C, Fantoni G, Candia J, Semba R, Ferrucci L. *A plasma proteomic signature of skeletal muscle mitochondrial function*. *International Journal of Molecular Sciences* 21:9540 (2020).
- 62) Tanaka T, Basisty N, Fantoni G, Candia J, Moore AZ, Biancotto A, Schilling B, Bandinelli S, Ferrucci L. *Plasma proteomic biomarker signature of age predicts health and life span*. *eLife* 9:e61073 (2020).
- 61) Candia J\*, Bayarsaikhan E\*, Tandon M\*, Budhu A, Forgues M, Tovuu L-O, Tudev U, Lack J, Chao A, Chinburen J, Wang XW. *The genomic landscape of Mongolian hepatocellular carcinoma*. *Nature Communications* 11:4383 (2020).
- 60) Liu J, Tang W, Budhu A, Forgues M, Hernandez MO, Candia J, Kim Y, Bowman ED, Ambs S, Zhao Y, Tran B, Wu X, Koh C, Surana P, Liang TJ, Guarnera M, Mann D, Rajaure M, Greten TF, Wang Z, Yu H, Wang XW. *A Viral Exposure Signature Defines Early Onset of Hepatocellular Carcinoma*. *Cell* 182:1-12 (2020).
- 59) Tanaka T, Lavery R, Varma V, Fantoni G, Colpo M, Thambisetty M, Candia J, Resnick SM, Bennett DA, Biancotto A, Bandinelli S, Ferrucci L. *Plasma proteomic signatures predict dementia and cognitive impairment*. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* e12018 (2020).
- 58) Osawa Y, Semba RD, Fantoni G, Candia J, Biancotto A, Tanaka T, Bandinelli S, Ferrucci L. *Plasma proteomic signature of the risk of developing mobility disability: A 9-year follow-up*. *Aging Cell* e13132 (2020).
- 57) Celik H, Lindblad K, Popescu B, Gui G, Goswami M, Valdez J, DeStefano C, Lai C, Thompson J, Ghannam J, Fantoni G, Biancotto A, Candia J, Cheung F, Sukumar G, Dalgard C, Smith R, Larochelle A, Dillon L, Hourigan C. *Highly multiplexed proteomic assessment of human bone marrow in acute myeloid leukemia*. *Blood Advances* 4:367-379 (2020).
- 56) Semba RD, Gonzalez-Freire M, Tanaka T, Biancotto A, Zhang P, Shardell M, Moaddel R, CHI Consortium, Ferrucci L. *Elevated Plasma Growth and Differentiation Factor 15 Is Associated With Slower Gait Speed and Lower Physical Performance in Healthy Community-Dwelling Adults*. *J Gerontol A Biol Sci Med Sci* 75:175–180 (2020).
- 55) Cheung F, CHI Consortium. *A Figure One Web Tool for Visualization of Experimental Designs*. *Journal of Open Research Software* 8:6 (2020).
- 54) Karmaus PWF, Shi M, Perl S, Biancotto A, Candia J, Cheung F, Kotliarov Y, Young NS, Fessler MB, CHI Consortium. *Effects of rosuvastatin on the immune system in healthy volunteers with normal serum cholesterol*. *JCI Insight* 4:e131530 (2019).
- 53) Lee BH, Kelly G, Bradford S, Davila M, Guo XV, Amir EAD, Thrash EM, Solga MD, Lannigan J, Sellers B, Candia J, Tsang J, Montgomery RR, Tamaki SJ, Sigdel TK, Sarwal MM, Lanier LL, Tian Y, Kim C, Hinz D, Peters B, Sette A, Rahman AH. *A Modified Injector and Sample Acquisition Protocol Can Improve Data Quality and Reduce Inter-Instrument Variability of the Helios Mass Cytometer*. *Cytometry Part A*, 95A:1019–1030 (2019).
- 52) Candia J, Tsang JS. *eNetXplorer: an R package for the quantitative exploration of elastic net families for generalized linear models*. *BMC Bioinformatics* 20:189 (2019).
- 51) Ciucci T, Vacchio MS, Gao Y, Tomassoni Ardori F, Candia J, Mehta M, Zhao Y, Tran B, Pepper M, Tessarollo L, McGavern DB, Bosselut R. *The emergence and functional fitness of memory CD4+ T cells require the transcription factor Thpok*. *Immunity* 50:91-105 (2019).
- 50) Brown D, Zingone A, Yu Y, Zhu B, Candia J, Cao L, Ryan BM. *Relationship between circulating inflammation proteins and lung cancer diagnosis in the National Lung Screening Trial*. *Cancer Epidemiology, Biomarkers & Prevention* 28(1):110-118 (2019).

- 49) Shen Y, Kubben N, Candia J, Morozov AV, Misteli T, Losert W. *RefCell: Multi-dimensional analysis of image-based high-throughput screens based on 'typical cells'*. BMC Bioinformatics 19:427 (2018).
- 48) Giudice V, Biancotto A, Wu Z, Cheung F, Candia J, Fantoni G, Kajigaya S, Rios O, Townsley D, Feng X, Young NS. *Aptamer-based proteomics of serum and plasma in acquired aplastic anemia*. Experimental Hematology 68:38-50 (2018).
- 47) Khurana S, Coyle EM, Manischewitz J, King LR, Gao J, Germain RN, Schwartzberg PL, Tsang JS, Golding H, CHI Consortium. *AS03-adjuvanted H5N1 vaccine promotes antibody diversity and affinity maturation, NAI titers, Cross-Clade H5N1 Neutralization, but not H1N1 cross-subtype neutralization*. npj Vaccines 3:40 (2018).
- 46) Chen J, Cheung F, Shi R, Zhou H, Lu W, CHI Consortium. *PBMC Fixation and Processing for Chromium Single-Cell RNA Sequencing*. Journal of Translational Medicine 16:198 (2018).
- 45) Tanaka T, Biancotto A, Moaddel R, Moore Z, Gonzalez-Freire M, Aon MA, Candia J, Zhang P, Cheung F, Fantoni G, CHI consortium, Semba RD, Ferrucci L. *Plasma proteomic signature of age in healthy humans*. Aging Cell e12799 (2018).
- 44) Candia J, Cheung F, Kotliarov Y, Fantoni G, Sellers B, Griesman T, Huang J, Stuccio S, Zingone A, Ryan BM, Tsang JS, Biancotto A. *Assessment of Variability in the SOMAscan Assay*. Scientific Reports 7:14248 (2017).
- 43) Cheung F, Fantoni G, Conner M, Sellers BA, Kotliarov Y, Candia J, Stagliano K, Biancotto A. *Web Tool for Navigating and Plotting SomaLogic ADAT Files*. Journal of Open Research Software 5:20 (2017).
- 42) Dolatabadi S, Candia J, Akrap N, Vannas C, Tomic TT, Losert W, Landberg G, Åman P, Ståhlberg A. *Cell cycle and cell size dependent gene expression reveals distinct subpopulations at single-cell level*. Frontiers in Genetics 8:1 (2017) [Reprinted in: Pan X, Wu S, Weissman SM (Eds) *Introduction to Single Cell Omics*. Lausanne: Frontiers Media. doi: 10.3389/978-2-88945-920-9, Chapter 3, p.108-118 (2019)].
- 41) Lau WW, Sparks R, OMiCC Jamboree Working Group, Tsang JS. *Meta-analysis of crowdsourced data compendia suggests pan-disease transcriptional signatures of autoimmunity*. F1000Research 5:2884(2016).
- 40) Chen D, Sarkar S, Candia J, Florczyk SJ, Bodhak S, Driscoll MK, Simon Jr CG, Dunkers JP, Losert W. *Machine learning based methodology to identify cell shape phenotypes associated with microenvironmental cues*. Biomaterials 104:104-118 (2016).
- 39) Candia J, Cherukuri S, Guo Y, Doshi KA, Banavar JR, Civin CI, Losert W. *Uncovering low dimensional, miR-based signatures of acute myeloid and lymphoblastic leukemias with a machine-learning-driven network approach*. Convergent Science Physical Oncology 1:025002 (2015).
- 38) Goswami M, McGowan KS, Lu K, Jain N, Candia J, Hensel NF, Tang J, Calvo KR, Battiwalla M, Barrett AJ, Hourigan CS. *A novel multi-gene array allows relapse risk stratification in acute myeloid leukemia patients undergoing stem cell transplantation*. Bone Marrow Transplantation 1-10 (2015).
- 37) Candia J, Banavar JR, Losert W. *Understanding Health and Disease with Multidimensional Single-Cell Methods*. J Phys: Condens Matter 26:073102 (2014).
- 36) Candia J, Maunu R, Driscoll M, Biancotto A, Dagur P, McCoy Jr JP, Sen HN, Wei L, Maritan A, Cao K, Nussenblatt RB, Banavar JR, Losert W. *From Cellular Characteristics to Disease Diagnosis: Uncovering Phenotypes with Supercells*. PLoS Computational Biology 9:e1003215 (2013).
- 35) Mazzitello KI, Candia J, Albano EV. *Far-from-equilibrium growth of magnetic thin films with Blume-Capel impurities*. Physical Review E 91:042118 (2015).
- 34) Baglietto G, Albano EV, Candia J. *Complex network structure of flocks in the Vicsek Model with Vectorial Noise*. Int J Mod Phys C 25:1350095 (2014).
- 33) Baglietto G, Albano EV, Candia J. *Complex network structure of flocks in the Standard Vicsek model*. J Stat Phys 153:270 (2013).
- 32) Baglietto G, Albano EV, Candia J. *Gregarious vs Individualistic Behavior in Vicsek Swarms and the Onset of First-Order Phase Transitions*. Physica A 392:3240 (2013).
- 31) Loscar ES, Candia J. *Stochastic Resonance and Dynamic First-Order Pseudo-Phase Transitions in the Irreversible Growth of Thin Films under Spatially Periodic Magnetic Fields*. Phys Rev E 88:042412(2013).

- 30) Candia J, Albano EV. *Nonequilibrium critical behavior of magnetic thin films grown in a temperature gradient*. J Stat Mech: Theor Exper P08006 (2012).
- 29) Baglietto G, Albano EV, Candia J. *Criticality and the onset of ordering in the Standard Vicsek Model*. Interface Focus 2:708 (2012).
- 28) Mazzitello K, Candia J. *Diffusion-based density-equalizing maps: an interdisciplinary approach to visualizing homicide rates and other georeferenced statistical data*. Braz J Phys 42: 365 (2012).
- 27) Candia J, Albano EV. *Far-from-equilibrium growth of thin films in a temperature gradient*. Phys Rev E 84:050601 (R) (2011).
- 26) Candia J. *Advertising and irreversible opinion spreading in complex social networks*. Int J Mod Phys C 20:799 (2009).
- 25) Candia J, Albano EV. *The magnetic Eden model*. Int J Mod Phys C 19:1617 (2008).
- 24) Candia J, Mazzitello K. *Mass media influence spreading in social networks with community structure*. J Stat Mech: Theor Exper P07007 (2008).
- 23) Parris PE, Candia J, Kenkre VM. *Random walk access times on partially-disordered complex networks: an effective medium theory*. Phys Rev E 77:061113 (2008).
- 22) Candia J, M.C. González, P. Wang, T. Schoenharl, G. Madey, A.-L. Barabási. *Uncovering individual and collective human dynamics from mobile phone records*. J Phys A 41:224015 (2008).
- 21) Candia J. *Nonequilibrium opinion spreading on 2D small-world networks*. J Stat Mech: Theor Exper P09001 (2007).
- 20) Candia J, Parris PE, Kenkre VM. *Transport properties of random walks on scale-free/regular-lattice hybrid networks*. J Stat Phys 129:323 (2007).
- 19) Mazzitello K, Candia J, Dossetti V. *Effects of mass media and cultural drift in a model for social influence*. Int J Mod Phys C 18:1475 (2007).
- 18) Candia J. *Irreversible opinion spreading on scale-free networks*. Phys Rev E 75:026110 (2007).
- 17) Candia J. *Irreversible growth of binary mixtures on small-world networks*. Phys Rev E 74:031101 (2006).
- 16) Manías V, Candia J, Albano EV. *Corner wetting in a far-from-equilibrium magnetic growth model*. Eur Phys J B 47:563 (2005).
- 15) Candia J. *Detectable neutrino fluxes due to enhanced cosmic ray densities in the Galactic Centre region*. J Cosmol Astrop Phys 11:002 (2005).
- 14) Beacom JF, Candia J. *Shower power: isolating the prompt atmospheric neutrino flux using electron neutrinos*. J Cosmol Astrop Phys 11:009 (2004).
- 13) Candia J, Roulet E. *Diffusion and drift of cosmic rays in highly turbulent magnetic fields*. J Cosmol Astrop Phys 10:007 (2004).
- 12) Candia J, Roulet E. *Rigidity dependent knee and cosmic ray induced high energy neutrino fluxes*. J Cosmol Astrop Phys 09:005 (2003).
- 11) Candia J, Mollerach S, Roulet E. *Cosmic ray spectrum and anisotropies from the knee to the second knee*. J Cosmol Astrop Phys 05:003 (2003).
- 10) Candia J, Albano EV. *Order-disorder criticality, wetting, and morphological phase transitions in the irreversible growth of far-from-equilibrium magnetic films*. J Magn Magn Mater 260:338 (2003).
- 9) Candia J, Roulet E, Epele LN. *Turbulent diffusion and drift in galactic magnetic fields and the explanation of the knee in the cosmic ray spectrum*. J High Energy Phys 12:033 (2002).
- 8) Candia J, Mollerach S, Roulet E. *Cosmic ray drift, the second knee and galactic anisotropies*. J High Energy Phys 12:032 (2002).
- 7) Candia J, Albano EV. *Quasi-wetting and morphological phase transitions in confined far-from-equilibrium magnetic thin films*. J Chem Phys 117:6699 (2002).
- 6) Candia J, Albano EV. *Interfacial phase transitions in a far-from-equilibrium magnetic growth model*. J Phys: Condens Matter 14:4927 (2002).



- 5) Candia J, Epele LN, Roulet E. *Cosmic ray photodisintegration and the knee of the spectrum*. *Astrop Phys* 17:23 (2002).
- 4) Candia J, Albano EV. *Irreversible growth of a binary mixture confined in a thin film geometry with competing walls*. *Phys Rev Lett* 88:016103 (2001).
- 3) Candia J, Albano EV. *Monte Carlo simulation of the irreversible growth of magnetic thin films*. *J Appl Phys* 90:5395 (2001).
- 2) Candia J, Albano EV. *Comparative study of an Eden model for the irreversible growth of spins and the equilibrium Ising model*. *Phys Rev E* 63:066127 (2001).
- 1) Candia J, Albano EV. *Non-equilibrium wetting transition in a magnetic Eden model*. *Eur Phys J B* 16:531 (2000).

## BOOK CHAPTERS

- 3) Candia J, SomaScan Bioinformatics: Normalization, Quality Control, and Assessment of Pre-Analytical Variation. In: Ruiz Romero C, Calamia V and Lourido L (Eds), *Protein Arrays: Methods and Applications*, Print ISBN 978-1-0716-4594-9, eBook ISBN 978-1-0716-4595-6, Springer Nature (2025), Chapter 9. DOI:10.1007/978-1-0716-4595-6\_9.
- 2) Candia J, Banavar J, Losert W (2015) *Uncovering Phenotypes with Supercells: Applications to Single-Cell Sequencing*. In: Wang X (Ed), *Single Cell Sequencing and Systems Immunology*, Translational Bioinformatics (Vol. 5), ISBN 978-94-017-9752-8, ISBN 978-94-017-9753-5 (eBook), Springer, Chapter 2.
- 1) Pawling A, Yan P, Candia J, Schoenharl T and Madey G (2010) *Anomaly detection in streaming sensor data*. In: Cuzzocrea A (Ed), *Intelligent Techniques for Warehousing and Mining Sensor Network Data*, DOI: 10.4018/978-1-60566-328-9 IGI Global – Hershey, PA.

## SOFTWARE

- 2) Candia J. *mutSigMapper*. R package available under GPLv3 license at <https://github.com/juliancandia/mutSigMapper>
- 1) Candia J, Tsang JS. *eNetExplorer*. R package available under GPLv3 license at <https://github.com/juliancandia/eNetExplorer>

## INVITED AND CONTRIBUTED TALKS (selected)

- *Exploring novel metrics of aging*. Invited speaker at the 2nd Biomarkers of Aging Symposium, NIA, Baltimore, 9/12/2024.
- *Assessment of Variability and Normalization Methods Using the Plasma 7k SomaScan Assay v4.1*. Keynote Speaker at the 2023 SomaLogic User Group Meeting, Boston, 5/23/2023.
- *Toward causality in clinical proteomics: advantages and challenges of highly specific tools*. Invited seminar at the 2022 HUPO World Congress, Cancún, 12/5/2022.
- *Assessment of Variability and Normalization Methods Using the Plasma 7k SomaScan Assay v4.1*. SomaLogic-sponsored webinar through GenomeWeb, 11/30/2022. Co-presented with Keenan Walker.
- *Computational Approaches to Characterize Alternative Splicing in Peripheral Arterial Disease*. 15th Annual Research on Aging Showcase, Johns Hopkins School of Medicine, 4/8/2022. Best presentation award winner (2nd place).
- *Serological response to human virome defines clinical outcome of Italian patients infected with SARS-CoV-2*. 17th Annual NCI Staff Scientists / Staff Clinicians Retreat (virtual), 4/30/2021. Outstanding Oral Presentation Award based on scientific merit and overall quality with \$1,500 travel award.
- *The genomic landscape of Mongolian hepatocellular carcinoma*. 3rd Annual Chinese-American Liver Society (CALS) Annual Symposium (virtual), 10/30/2020. Best presentation award.

- *Nonlinear Mixed-Effects Modeling of Proteomics Antibody-Based Multiplex Assays: A Bioinformatics Post-Hoc Approach to Improve Signal-to-Noise Ratios*. ISMB/ECCB Conference, Prague, 07/25/2017.
- *Meta-Analysis of CHI's SOMAscan Data: Assessing Variability, Reproducibility, and Reusability*. 1<sup>st</sup> SOMAscan Users Workshop. Center for Human Immunology, NIH, 03/27/2017.
- *Flow Cytometry Bioinformatics*. Notable Labs, 01/11/2017.
- *Exploring Protein Biomarkers with the SOMAscan Assay: From Wet-Lab to Bioinformatics*. Icahn School of Medicine at Mt. Sinai, 8/24/2016.
- *SomaLogic SOMAscan proteomics assay* (co-presented with A. Biancotto) in the “Genomics and Immunology Science Brown Bag Series”, Clinical Genomics Program, NIAID, NIH, 3/09/2016.
- *Machine learning algorithms for flow cytometry analysis of human disease*. Institute of Inflammation and Ageing Seminar, University of Birmingham (UK), 11/17/2015.
- *Unbiased Learning from Big Data: Multidimensional Approaches for Data-Driven Biomedical Research*. Division of Biostatistics and Bioinformatics, University of Maryland (Baltimore), 06/24/14.
- *Uncovering Differential Multi-microRNA Signatures of Acute Myeloid and Lymphoblastic Leukemias with a Machine-Learning-Based Network Approach*. “2014 Joint Summits on Translational Science”, American Medical Informatics Association (AMIA), 04/07-11/14, San Francisco.
- *From molecules to cells to organisms: understanding health and disease with multidimensional single-cell methods*. Invited Talk at the American Physical Society March Meeting, Baltimore, 03/21/2013.
- *Solving the Puzzle of Cell Heterogeneity vs Disease Phenotype with Supercells*. Invited talk at the UMD-NCI Cancer Technology Workshop, UMD, College Park, 01/18/2013.
- *Cell Averaging and Classification of Multidimensional Single-Cell Data: Solving the Puzzle of Heterogeneity vs Disease Phenotype with Supercells*. Pacific Symposium on Biocomputing 2013, Kona, Hawaii, 01/03/2013.
- *Uncovering Cell Subpopulations in Health and Disease: a Network Analysis of Multidimensional Single-Cell Data*. FOCIS-CHI Lecture Series “Bedside to Bench and Back”, NIH, Bethesda, 03/21/2012.

## AWARDS

- Scientific/Medical NIA Director's Award (2024).
- Length of Service Award for 10 years of Federal service (2024).
- Performance awards obtained every year since joining NIH (2014-2023).
- NIA Quadrennial performance review graded “Outstanding to Exceptional” for the period 2018-2022.
- NIAID Quadrennial performance review graded “Outstanding” for the period 2014-2018.
- Award from the NIH T32 Training Program in Cancer Biology (2012-2014).
- Awards to perform research at Fermilab, ICTP, UNM, Notre Dame, Northeastern, UMD.
- Award from CONICET (Argentina) to perform research for a Ph.D. degree (2000-2004). This was one out of approximately 250 awards granted to candidates in all Ph.D. programs in Sciences, Technology, and Humanities in Argentina.
- Awards from Fundación Antorchas (Argentina) to perform research in undergraduate (1998-1999), graduate (2000) and postdoc (2004) roles.
- “Dr. J.V. González” Award for highest GPA from the University of La Plata (2000).

## MENTORING

- Mentor for NIH OITE Summer Internship Program (2022). Mentee: Sachin Kammula (undergraduate student at Johns Hopkins University, Baltimore).
- Mentor for UMD-NCI Partnership for Integrative Cancer Research (2019-2021). Mentee: Zeynep Kacar (Ph.D. Candidate at the University of Maryland, College Park).

- Mentor for NIH OITE Summer Internship Program (2019). Mentee: Ashwin Kammula (undergraduate student at the University of Maryland, College Park).
- Mentor for NIH Data Science Mentoring Program (2017). Mentee: Dr. Thomas Ciucci (Postdoctoral Fellow at Remy Bosselut's Lab, LICB, NCI).

#### **EDITOR AND REVIEWER SERVICE**

- Editor for Scientific Reports, Nature Portfolio (since 2023).
- Editor for Clinical and Experimental Medicine, Springer Nature (since 2019).
- Reviewer for Bioinformatics, J. Gerontol.: Biol. Sci., Clin. Chem., JNCI, PLoS One, J. Transl. Med., Cytometry Part A, J. Royal Soc. Interface, Nature Comm., AMIA Transl. Bioinformatics, AMIA Joint Summits on Transl. Science, Clin. Cancer Res., Comm. Biol., J. Proteom., Cell & Bioscience, and others.
- Grant reviewer for Hartstichting—Dutch Heart Foundation (2023).
- NIH Stadtman Search Computational Biology/Bioinformatics/Biostatistics/Mathematics Committee member (2021, 2022, 2023, 2024).

#### **LANGUAGES**

- Fluent in English, Spanish, and Italian.